

Written examination in the course

Engineering Optimisation - Concepts, Methods and Applications

Tuesday June 11th 2013

kl. 9.00 - 13.00 (4 hours)

All usual helping aids are allowed, i.e. books, notes, calculator, computer etc. All communication equipment and computer communication protocols must be turned off.

You may answer the questions in English or Danish.

REMEMBER to write your study number and page number on all sheets handed in.

The set consists of 7 exercises. The total weighting for each of the exercises is stated in percentage. You need 50 % in order to pass the exam.

It should be clear from the solution, which methods are used, and there should be a sufficient number of intermediate calculations, so the line of thought is clear.

Exercise 1: (10 %)

The following optimisation problem is considered:

$$\begin{aligned} \text{Minimise } f(\mathbf{x}) &= (2 - x_1)^2 + (x_2 + 1)^2 \\ \text{Subject to } h(\mathbf{x}) &= -x_1 - x_2 + 4 = 0 \end{aligned} \quad (1)$$

- Set up the Lagrangian function and find point(s) satisfying the KKT necessary conditions.
- Check if the point(s) is an optimum point using the graphical method (make a simple sketch).

Exercise 2: (15 %)

We will consider gradient-based minimisation of the following unconstrained function:

$$f(\mathbf{x}) = 10(x_1^2 - x_2) + x_1^2 - 2x_1 + 5 \quad (2)$$

The starting point is: $\mathbf{x}^{(0)} = [-1 \ 3]^T$.

- Complete the first iteration of the steepest descent method for the function. The 1D line search problem should be solved analytically.
- Can Newton's method be applied for determining the search direction in iteration 1? If yes, then determine the search direction. If no, then state an alternative robust method for determining the search direction.

Exercise 3: (18 %)

An optimisation problem is given as:

$$\text{minimise } f(\mathbf{x}) = (x_1 - 1)^2 + 2(x_2 - 1)^2 - x_1x_2 \quad (3)$$

Subject to the constraints:

$$\begin{aligned} g_1(\mathbf{x}) &= 5 - x_1 - x_2 && \leq 0 \\ g_2(\mathbf{x}) &= x_1^2 - x_2 - 36 && \leq 0 \\ g_3(\mathbf{x}) &= \frac{x_1^2}{4} - x_1 + x_2 - 8 && \leq 0 \\ g_4(\mathbf{x}) &= -x_1 && \leq 0 \\ g_5(\mathbf{x}) &= -x_2 && \leq 0 \end{aligned}$$

Handwritten note: *Handwritten step 4* (4)

a) Complete one iteration of the Sequential Linear Programming (SLP) method for the above problem, where you solve step 4 graphically. Use a starting point of $\mathbf{x}^{(0)} = (5, 5)$ and 20% move limits. As a help the contour plot of the linearised objective function for $\mathbf{x}^{(0)} = (5, 5)$ is shown in figure 1. The page may be handed in with the solution.

b) Describe in words, if there are any limitations in using the SLP-method and/or if there are any type of problems, for which the SLP-method is unsuited.

Exercise 4: (7 %)

The following multi-objective optimisation problem is considered:

$$\begin{aligned} \text{minimise} \quad & f_1(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 2)^2 + 1 \\ & f_2(\mathbf{x}) = (x_1 - 6)^2 + (x_2 - 6)^2 + 4 \end{aligned} \quad (5)$$

Figure 2 shows the Pareto optimal points in the design space and figure 3 shows the Pareto optimal set in the criterion space.

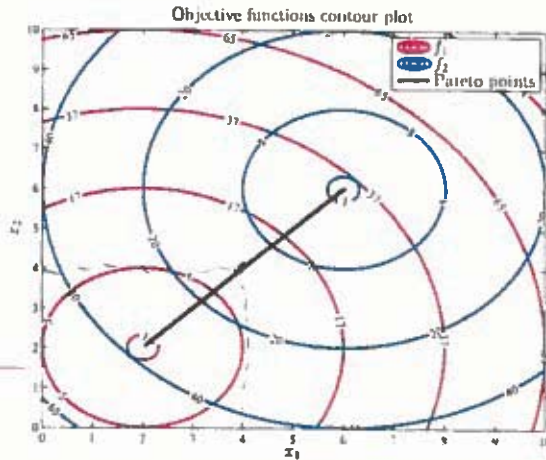


Figure 2: Contour curves and Pareto optimal points for the problem of exercise 4.

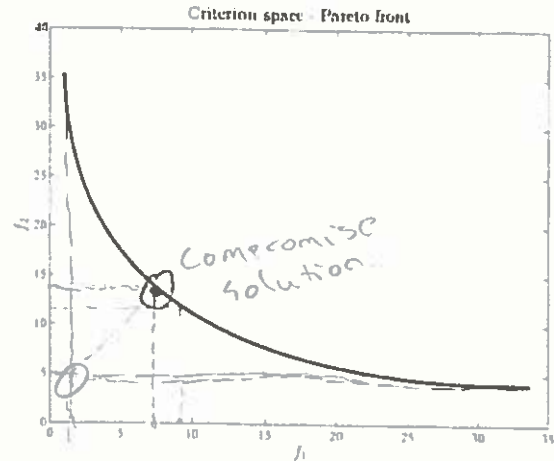


Figure 3: Pareto set in criterion space for the problem of exercise 4.

- Determine the objective function values of the utopia point.
- Assume that the multi-objective problem is solved as a single objective problem using the weighting method (i.e. minimise $U(\mathbf{x}) = w_1 f_1(\mathbf{x}) + w_2 f_2(\mathbf{x})$), with $w_1 = w_2 = 1$. Determine the minimum objective function value $U(\mathbf{x}^*)$, and the optimum set of design variables $\mathbf{x}^*$.

Exercise 5: (20 %)

The cost function $f(\mathbf{x})$ is defined as the sum of squares of the residual function $\tau(\mathbf{x})$

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^m r_i^2 \quad (6)$$

where

$$r_i(\mathbf{x}) = \sqrt{(x_1 - p_i)^2 + (x_2 - q_i)^2} - y_i \quad (7)$$

Note that the problem only holds two variables $\mathbf{x} = [x_1 \ x_2]$ ($n = 2$) and five points in the residual vector ($m = 5$). The parameters (constants or vectors holding five components each) p_i , q_i and y_i are given by

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{bmatrix} = \begin{bmatrix} 1.8 \\ 2.0 \\ 1.5 \\ 1.5 \\ 2.5 \end{bmatrix}, \quad \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 1.7 \\ 1.5 \\ 2.0 \\ 1.5 \end{bmatrix}, \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 1.87 \\ 1.24 \\ 0.53 \\ 1.29 \\ 1.49 \end{bmatrix} \quad (8)$$

The initial cost function $f(x_0) \approx 3.5937$ for $x_0 = [1.5, 4]$. The minimum value $f(x^*) \approx 0.03050167$ at $x^* \approx [1.1833, 0.8275]$, see figure 4.

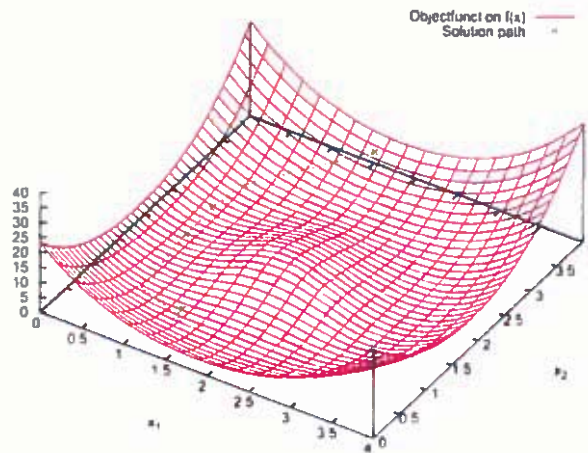


Figure 4: Surface plot of the cost function and the trajectory produced by an iterative solution scheme.

The following questions must be answered:

1. Derive an analytically expression for the Jacobian matrix associated with equation 7.
2. Numerical solutions are expected for the following three sub-questions, where the initial parameters $x_0 = [1.5, 4]$ are inserted into the expressions derived above.
 - (a) Using the initial parameters $x_0 = [1.5, 4]$; calculate the gradient and approximate the Hessian using only first order information from the Jacobian matrix derived above in question 1.
 - (b) Complete the first iteration using an unscaled (pure) Newton step and the initial parameters $x_0 = [1.5, 4]$.
 - (c) Construct a Taylor series representing the non-linear function $f(x_0)$ where $x_0 = [1.5, 4]$ using the gradient and Hessian calculated in question 2a. Use the Taylor series to approximate the function value at the point $x_1 = x_0 + s$ where s is the pure Newton step from question 2b.
3. Is the Newton direction always that of descent of the cost function? Explain why or why not.

Exercise 6: (15 %)

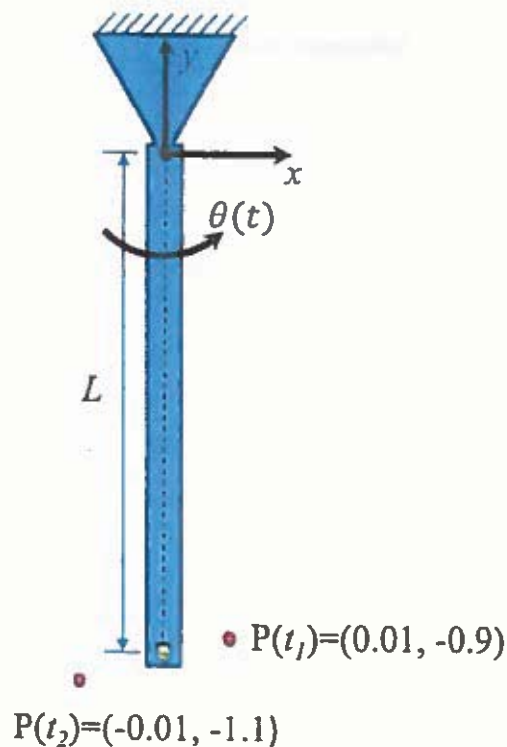


Figure 5: A rigid body connected to ground through a revolute joint.

Figure 5 shows a rigid body that is connected to ground through a revolute joint in a two-dimensional space. The body is assumed to be infinitely thin. A marker (the green dot in the figure) is attached to the body a distance L from the rotation point of the revolute joint. The global reference frame is indicated with axes x and y . The rotation angle of the body is denoted $\theta(t)$ and is measured between the long axis of the body and the vertical axis, i.e. the global y -axis.

A motion capture experiment was performed, where the position of the marker was measured in the global reference frame at two time instances, $t_1 = 0$ s and $t_2 = 1$ s to be $P(t_1) = (0.01 \text{ m}, -0.9 \text{ m})$ and $P(t_2) = (-0.01 \text{ m}, -1.1 \text{ m})$. From the measured positions of the marker, it is desired to compute the orientation of the rigid body, $\theta(t_1)$ and $\theta(t_2)$ as well as the optimal length, L , such that the sum of the least-square differences between the measured marker and the point on the body is minimised for both time steps.

The following questions must be answered:

1. Is the system of kinematic equations under-determinate, determinate or over-determinate when the motion of the rigid body connected to ground with a revolute joint is driven by the measured marker? Justify the answer by counting the number of degrees-of-freedom of the mechanical system compared to the number of kinematic driver equations.
2. Formulate a least-squares optimisation problem that can be used to both compute the optimal length, L , and the angles $\theta(t_1)$ and $\theta(t_2)$, simultaneously.
3. Derive the first-order optimality conditions for the optimisation problem derived in question 2. NB! You do not have to solve the equations.

Exercise 7: (15 %)

We assume you are going to solve a design optimisation problem where the maximum pressure of fluid running in a network of pipes is to be minimised.

The pipe system consists of $n = 10$ pipes, the design variables of the problem are denoted x_i , $i = 1, \dots, n$, and these are the cross sectional areas of the pipes. We assume that you have been given an analysis model that can compute the pressure distribution in the network of pipes, and this will give you pressure values P_i , $i = 1, \dots, n$, related to the pipes.

- a) Explain the main mathematical difficulty of solving the problem of minimising the maximum pressure in the n pipes.
- b) State the mathematical formulation for three fundamentally different ways of solving the optimisation problem. Describe advantages and disadvantages of the three chosen optimisation problem formulations.

Exercise 1

The following optimisation problem is considered

$$\text{Minimize: } f(x) = (2-x_1)^2 + (x_2+1)^2$$

$$\text{st. } h(x) = -x_1 - x_2 + 4 = 0$$

a) set up the Lagrangian function and find point(s) satisfying the KKT necessary conditions

$$L(x, v, u, s) = f(x) + \underbrace{\sum_{j=1}^p v_j h_j(x)}_{\text{equality constraints}} + \underbrace{\sum_{i=1}^m u_i (g_i(x) + s_i^2)}_{\text{inequality constraints}}$$

Lagrangian function

Plug in $\rightarrow L(x, v) = (2-x_1)^2 + (x_2+1)^2 + v_1 \cdot (-x_1 - x_2 + 4)$

necessary conditions

$$\frac{\partial L}{\partial x_1} = 2 \cdot x_1 - v_1 - 4 = 0$$

$$\frac{\partial L}{\partial x_2} = 2 \cdot x_2 - v_1 + 2 = 0$$

$$\frac{\partial L}{\partial v_1} = -x_1 - x_2 + 4 = 0$$

Solve

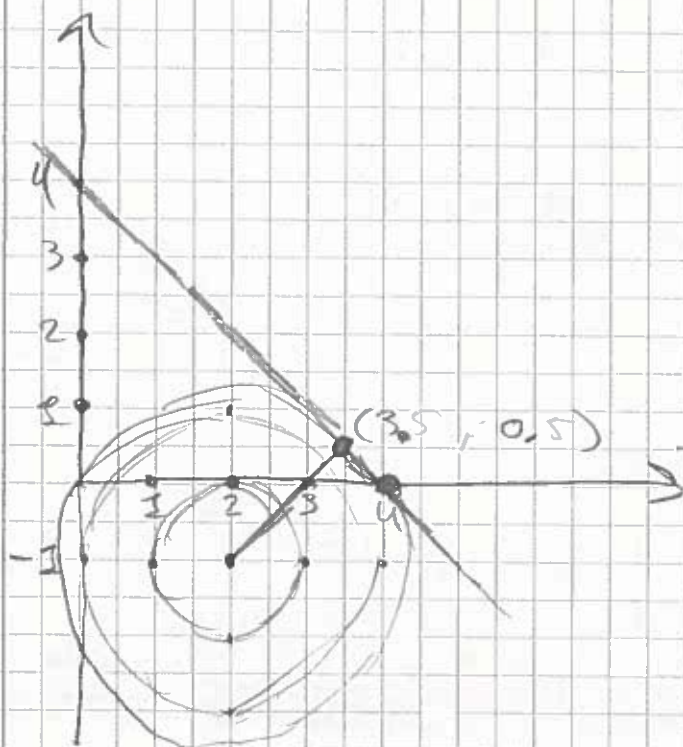
together

$$v_1 = 3$$

$$x_1 = 3.5$$

$$x_2 = 0.5$$

b) check if the point(s) is an optimum point using graphical method (make a simple sketch).



it is optimum

$$f(3.5, 0.5) = 4.5$$

We will consider gradient-based minimization of the following unconstrained function.

$$f(x) = 10(x_1^2 - x_2) + x_1^2 - 2x_1 + 5$$

The starting point is $x^{(0)} = [-1, 3]^T$

a) Complete the first iteration of the steepest descent method for the function. The 1D line search problem should be solved analytically.

Step 1 function value at starting point

$$f(x^{(0)}) = -12$$

Step 2 Calculate the gradient at the starting point.

$$c(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 22x_1 - 2 \\ -10 \end{bmatrix} \quad c(x^{(0)}) = \begin{bmatrix} -24 \\ -10 \end{bmatrix}$$

Step 3 norm. of the gradient

$$\|c^{(0)}\| = \sqrt{(-24)^2 + (-10)^2} = 26 > \epsilon$$

Step 4 Direction of steepest descent is found from the gradient

$$d^{(0)} = -c(x^{(0)}) = -c^{(0)} = \begin{bmatrix} 24 \\ 10 \end{bmatrix}$$

Step 5 Compute step size using 1D (line search)

$$f(\alpha) = f(x^{(0)} + \alpha d^{(0)})$$

$$= 10 \cdot ((-1 + 24\alpha)^2 - (3 + 10\alpha)) + (-1 + 24\alpha)^2 - 2(-1 + 24\alpha) + 5$$

• using that: $x^{(0)} + \alpha d^{(0)} = \begin{bmatrix} -1 + 24\alpha \\ 3 + 10\alpha \end{bmatrix}$

• Differentiate to find the step size

$$\frac{df(\alpha)}{d\alpha} = 12672\alpha - 676 = 0 \Rightarrow \alpha = \frac{676}{12672} = \underline{0.053}$$

Step 6 Update design point

$$x^{(1)} = x^{(0)} + \alpha d^{(0)} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} + 0.0533 \begin{bmatrix} 24 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.28 \\ 3.53 \end{bmatrix}$$

Can Newton's method be applied to determine search direction in iteration 2? if yes do so. If no stated calculate the Hessian to check for singularity.

$$H = \begin{bmatrix} 22 & 0 \\ 0 & 0 \end{bmatrix}$$

we need the H^{-1} for the Newton's method but that gives singularity $\Rightarrow$ N/A

Alternative 2 Conjugate gradient

should always be used instead of steepest descent.

$$d^{(k)} = -\nabla f(x^{(k)}) + \beta_k d^{(k-1)}$$

$$\beta_k = \left(\frac{\| \nabla f(x^{(k)}) \|^2}{\| \nabla f(x^{(k-1)}) \|^2} \right)$$

uses only gradients so less computation heavy and converges faster.

An optimization problem is given as

$$\text{Minimize } f(x) = (x_1 - 1)^2 + 2(x_2 - 1)^2 - x_1 x_2$$

$$\text{s.t. } g_1(x) = 5 - x_1 - x_2 \leq 0$$

$$g_2(x) = x_1^2 - x_2 - 36 \leq 0$$

$$g_3(x) = \frac{x_1^2}{4} - x_1 + x_2 - 8 \leq 0$$

$$g_4(x) = -x_1 \leq 0$$

$$g_5(x) = -x_2 \leq 0$$

a) Complete one iteration of Sequential linear programming (SLP) method for the above problem, where you solve step 4 graphically. Use a starting point of $x^{(0)} = (5, 5)$ and 20% move limit.

Normalize the constraints

$$g_1(x) = -\frac{x_1}{5} - \frac{x_2}{5} + 1 \leq 0$$

$$g_2(x) = \frac{x_1^2}{36} - \frac{x_2}{36} - 1 \leq 0$$

$$g_3(x) = \frac{x_1^2}{32} - \frac{x_1}{8} + \frac{x_2}{8} - 1 \leq 0$$

$$g_4(x) = -x_1 \leq 0$$

$$g_5(x) = -x_2 \leq 0$$

Step 1 $x^{(0)} = (5, 5)$

Step 2

Cost function and constraints at starting point

$$f(5, 5) = 23$$

$$\left. \begin{aligned} g_1(5, 5) &= -1 \\ g_2(5, 5) &= -\frac{4}{9} \\ g_3(5, 5) &= -\frac{23}{32} \\ g_4(5, 5) &= -5 \\ g_5(5, 5) &= -5 \end{aligned} \right\} \text{inactive}$$

gradients of cost and constraints

$$\nabla f = \begin{bmatrix} 2x_1 - x_2 - 2 \\ -x_1 + 4x_2 - 4 \end{bmatrix}; \nabla f(5, 5) = \begin{bmatrix} 3 \\ 11 \end{bmatrix}$$

$$\nabla g_1 = \begin{bmatrix} -\frac{1}{5} \\ -\frac{1}{5} \end{bmatrix}; \nabla g_2 = \begin{bmatrix} \frac{x_1}{18} \\ -\frac{1}{36} \end{bmatrix}; \nabla g_2(5, 5) = \begin{bmatrix} \frac{5}{18} \\ -\frac{1}{36} \end{bmatrix}$$

$$\nabla g_3 = \begin{bmatrix} \frac{x_1}{16} - \frac{1}{8} \\ \frac{1}{8} \end{bmatrix}; \nabla g_3(5, 5) = \begin{bmatrix} \frac{3}{16} \\ \frac{1}{8} \end{bmatrix}$$

$$\nabla g_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}; \nabla g_5 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Move limits: 20%

$$-0.25 = -1 \leq d_1 \leq 0.25 = 1$$

$$-0.25 = -1 \leq d_2 \leq 0.25 = 1$$

as no equality constraint $\Rightarrow$ no bound

The A-matrix: $A = \begin{bmatrix} \nabla g_1 & \nabla g_2 & \dots & \nabla g_5 \end{bmatrix}$

$$= \begin{bmatrix} -1/5 & 5/8 & 3/16 & -1 & 0 \\ -1/5 & -1/36 & 1/8 & 0 & -1 \end{bmatrix}$$

next Define the Linear Programming (LP) sub-problem

minimize $\bar{f} = c^T d$
st. $N^T d = e$
 $A^T d \leq b$

The b vector: $b = \begin{bmatrix} -g_1(x^{(0)}) \\ -g_2(x^{(0)}) \\ -g_3(x^{(0)}) \\ -g_4(x^{(0)}) \\ -g_5(x^{(0)}) \end{bmatrix} = \begin{bmatrix} 1 \\ 4/9 \\ 7/32 \\ 5 \\ 5 \end{bmatrix}$

The subproblem is then

minimize $\bar{f} = c^T d \Rightarrow [3 \ 1] \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = 3d_1 + 1d_2$

s.t. $A^T d \leq b \Rightarrow \begin{bmatrix} -1/5 & -1/5 \\ 5/8 & -1/36 \\ 3/16 & 1/8 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} 1 \\ 4/9 \\ 7/32 \\ 5 \\ 5 \end{bmatrix}$

step 4

Solve for $d \rightarrow$ using SLP code

$$\Rightarrow d_1 = -1 \text{ and } d_2 = -1$$

step 5

check for convergence No stopping criteria.

norm of d $\|d^{(0)}\| = \sqrt{1^2 + 1^2} = \sqrt{2} \geq \epsilon_2$ not converged.

check constraints $g_1(-1,1) = 1$ $g_2(-1,1) = -1$ $g_3(-1,1) = -\frac{23}{32}$
 $g_4(-1,1) = 5$ $g_5(-1,1) = -1$

step 6 updating design

$$x^{(1)} = x^{(0)} + d = \begin{bmatrix} 5 \\ 5 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Limitations/problems with SLP?

Linearized problems may become unbounded/oscillating $\Rightarrow$ imposing move limits

Regarding move limits: May lead to non-converging if wrong settings $\Rightarrow$ set to a fraction of current design value

Cannot be used as black box approach $\Rightarrow$ move limit

Exercise

The following multi objective optimisation problem is considered

$$\text{minimize } f_1(x) = (x_1 - 2)^2 + (x_2 - 2)^2 + 1$$

$$f_2(x) = (x_1 - 6)^2 + (x_2 - 6)^2 + 1$$

Figure 2 shows Pareto optimal points in design space. Figure 3 shows Pareto optimal set in criterion space.

a) Determine the objective function values of the utopia point.

Point	f_1	f_2
$P_1(2, 2)$	1	36
$P_2(6, 6)$	33	1

$$u_{\text{topia}} = (\min(1, 33), \min(36, 1))$$
$$= \underline{(1, 1)}$$

b) $U(x) = w_1 f_1(x) + w_2 f_2(x)$ with $w_1 = w_2 = 1$

Determine min value of U and corresponding x^*

The compromise solution with lowest U is located at a point on the Pareto front that is located the shortest distance from the Utopia point.

That is also approx midway between the Pareto points in design space

$$x^* = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$U = f_1(4, 4) + f_2(4, 4)$$
$$= 9 + 12 = 21$$

The cost function $f(x)$ is defined as the sum of squares of the residual function $r(x)$

$$f(x) = \frac{1}{2} \sum_{i=1}^m r_i^2$$

where

$$r_i(x) = \sqrt{(x_1 - p_i)^2 + (x_2 - q_i)^2} - y_i$$

Note that the problem only holds two variables $x = [x_1, x_2]^T$ ($n=2$) and five points in the residual vector ($m=5$). The parameters (constants or vectors holding five components each) p_i , q_i , and y_i are given by

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{bmatrix} = \begin{bmatrix} 1.8 \\ 2.0 \\ .5 \\ 1.5 \\ 2.5 \end{bmatrix}, \quad \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 1.7 \\ 1.5 \\ 2.0 \\ 1.5 \end{bmatrix}, \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 1.87 \\ 1.24 \\ 0.53 \\ 1.29 \\ 1.44 \end{bmatrix}$$

The initial cost function $f(x_0) \approx 3.5937$ for $x_0 = [1.5, 4]^T$

The minimum value $f(x^*) \approx 0.03050167$ at $x^* \approx [1.1833, 0.8275]^T$

1/ Derive analytically expression for the jacobian.

$$J = \begin{bmatrix} \frac{\partial r_1}{\partial x_1} & \frac{\partial r_1}{\partial x_2} \\ \frac{\partial r_2}{\partial x_1} & \frac{\partial r_2}{\partial x_2} \\ \frac{\partial r_3}{\partial x_1} & \frac{\partial r_3}{\partial x_2} \\ \frac{\partial r_4}{\partial x_1} & \frac{\partial r_4}{\partial x_2} \\ \frac{\partial r_5}{\partial x_1} & \frac{\partial r_5}{\partial x_2} \end{bmatrix} = \begin{bmatrix} x_1 - p_1 & \sqrt{x_1^2 - 2x_1p_1 + p_1^2 + q_1^2 - 2q_1x_2 + x_2^2} \\ \vdots & \vdots \\ x_2 - q_1 & \sqrt{x_2^2 - 2x_2q_1 + p_1^2 - 2p_1x_1 + q_1^2 + x_1^2} \\ \vdots & \vdots \end{bmatrix}$$

2/ Numerical solutions expected. $x_0 = [1.5, 4]^T$

a) using x_0 ; calculate gradient and Hessian using only 1st order info.

$$\text{Gradient: } \nabla f(x) = J^T \cdot r = \begin{bmatrix} -0.0978 \\ 1.2804 \end{bmatrix}$$

$$\text{Hessian: } H = J^T \cdot J = \begin{bmatrix} 0.2215 & -0.7447 \\ -0.7447 & 4.7785 \end{bmatrix}$$

b) First iteration of pure newton step.

$$S_{\text{gauss}} = -H_{\text{gauss}}^{-1} \cdot \nabla f = \begin{bmatrix} -0.8801 \\ -1.0896 \end{bmatrix}$$

$$x^1 = x^0 + S_{\text{gauss}}$$

